

BioVerify

Early Flu Detection for Schools

A low-cost biosensing system that detects influenza signals in school air and provides verified early warnings to administrators.



Biology



Hardware



AI Verification

The Problem

Flu Is a Global Health & Economic Crisis

GLOBAL HEALTH BURDEN



WHO

~1B

people infected
with flu every year



WHO

650K

deaths annually from
flu complications



CDC

51M

flu illnesses in the
U.S. in 2024–25 alone

ECONOMIC IMPACT



WHO

\$60B

global economic cost
from flu per year



CDC

\$29B

cost to U.S. economy
in one flu season



Study

~65%

of sick kids miss school;
~53% of parents miss work

Traditional flu surveillance relies on lab-confirmed cases, which typically lag 1-2 weeks behind actual transmission – CDC Flu View

Our Solution

Introducing BioVerify

BioVerify continuously monitors school air for influenza markers and sends verified early alerts to administrators before outbreaks begin.

**Real-time biosensing → AI verification →
Targeted school response**



Air Sampling

Devices sample air in high-traffic areas



Biosensor Detection

Strip detects influenza viral markers



Verified Alert

AI verifies signal, alerts administrator

How It Works

Air Biosensing Device



Air Inlet

Fan pulls classroom air in



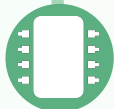
Capture Membrane

Moist strip traps viral particles



Flu Detection Strip

Biosensing chemistry reacts to influenza markers



PCB Signal Reader

Optical/electrical reader processes result



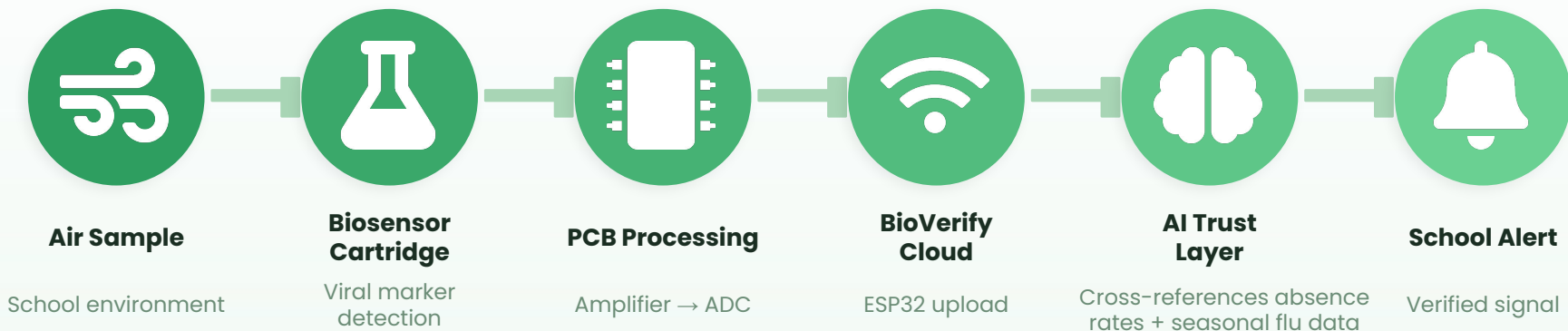
ESP32 Transmission

Wireless signal sent to cloud

Device Specifications

Microcontroller	ESP32-S3
Detection Method	Antigen / RNA strip
Air Sampling	Micro fan + capture membrane
Signal Output	Optical color-change reader
Connectivity	WiFi to cloud dashboard
Target Cost	< \$100 per unit
Target Pathogen	Influenza A & B

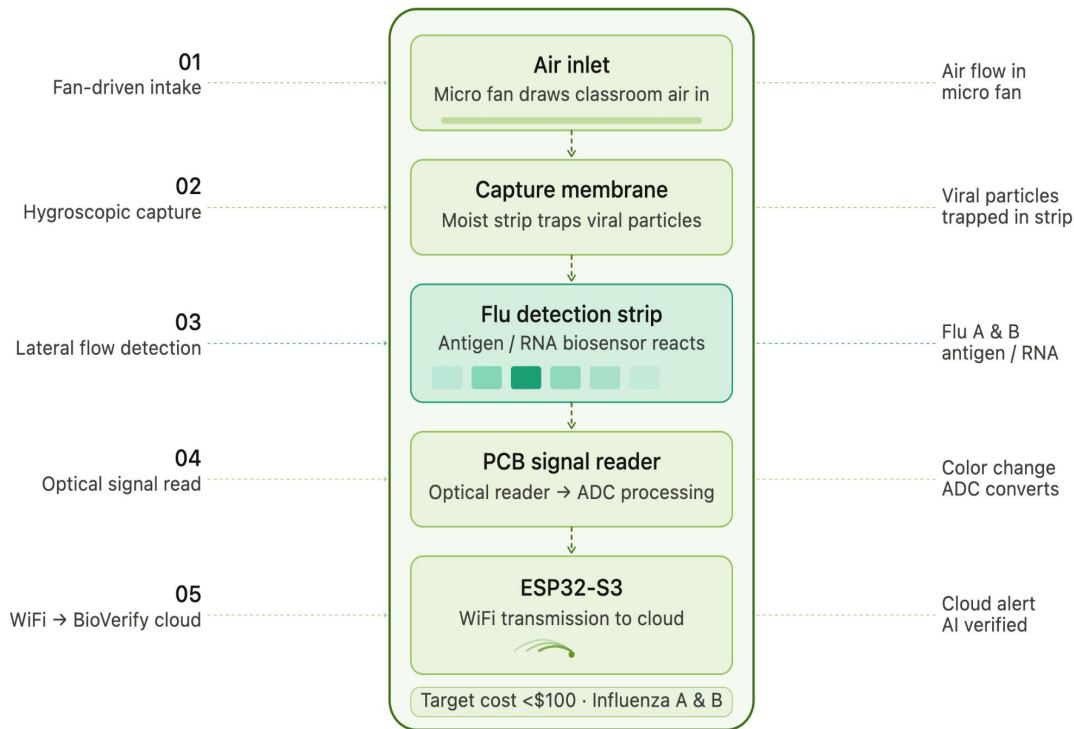
From Air Sample to School Alert



PCB reads biosensor signal → **ESP32** transmits wirelessly → **AI layer** verifies before alert is sent → No false positives reach administrators

Prototype

Inside The Device



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The Science Works

Influenza A&B shed **detectable viral RNA** in exhaled air

Lateral flow antigen strips validated for **airborne particle biosensing** in clinical settings

ESP32 microcontrollers used in **peer-reviewed** environmental biosensing research

(Fabian et al., 2008 — Nature)

(Posthuma-Trumpie et al., 2009)

(Andrade et al., 2021)

BioVerify Monitoring Dashboard

ZONE STATUS

Hallway A	Normal
Classroom B	Normal
Cafeteria	Elevated
Gym	Normal
Library	Normal
Science Lab	Alert
Main Office	Normal

NETWORK

12
Sensors
online

2
Active alerts

Last sync 6s ago

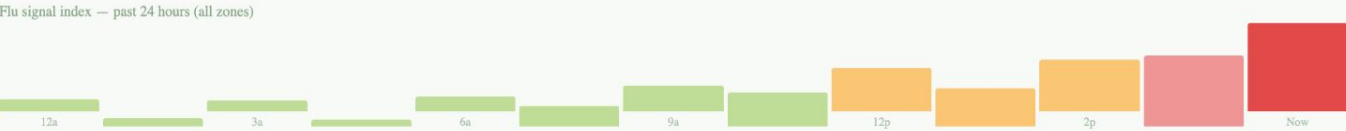
▲

Influenza A signal detected
Cafeteria sensor C-04 · Confidence: 92% · 14 min ago

ELEVATED

Recommended actions

- ✓ Increase ventilation in cafeteria immediately
- ✓ Monitor symptoms in zones C and D
- ✓ Schedule deep clean post-lunch period
- ✓ Notify school nurse — follow-up protocol active



LIVE READINGS

Cafeteria C-04

Elevated

0.74 index

Threshold 0.60 · Triggered 14 min ago

Science Lab S-02

Alert

0.81 index

Threshold 0.60 · Triggered 3 min ago

Hallway A-01

Normal

0.12 index

Threshold 0.60 · Last read 8s ago

One School. One Sensor. System-Wide Impact

300 000

infections prevented

annually, scaled across 10,000 schools

\$120M+

reduction

in direct healthcare costs per flu season

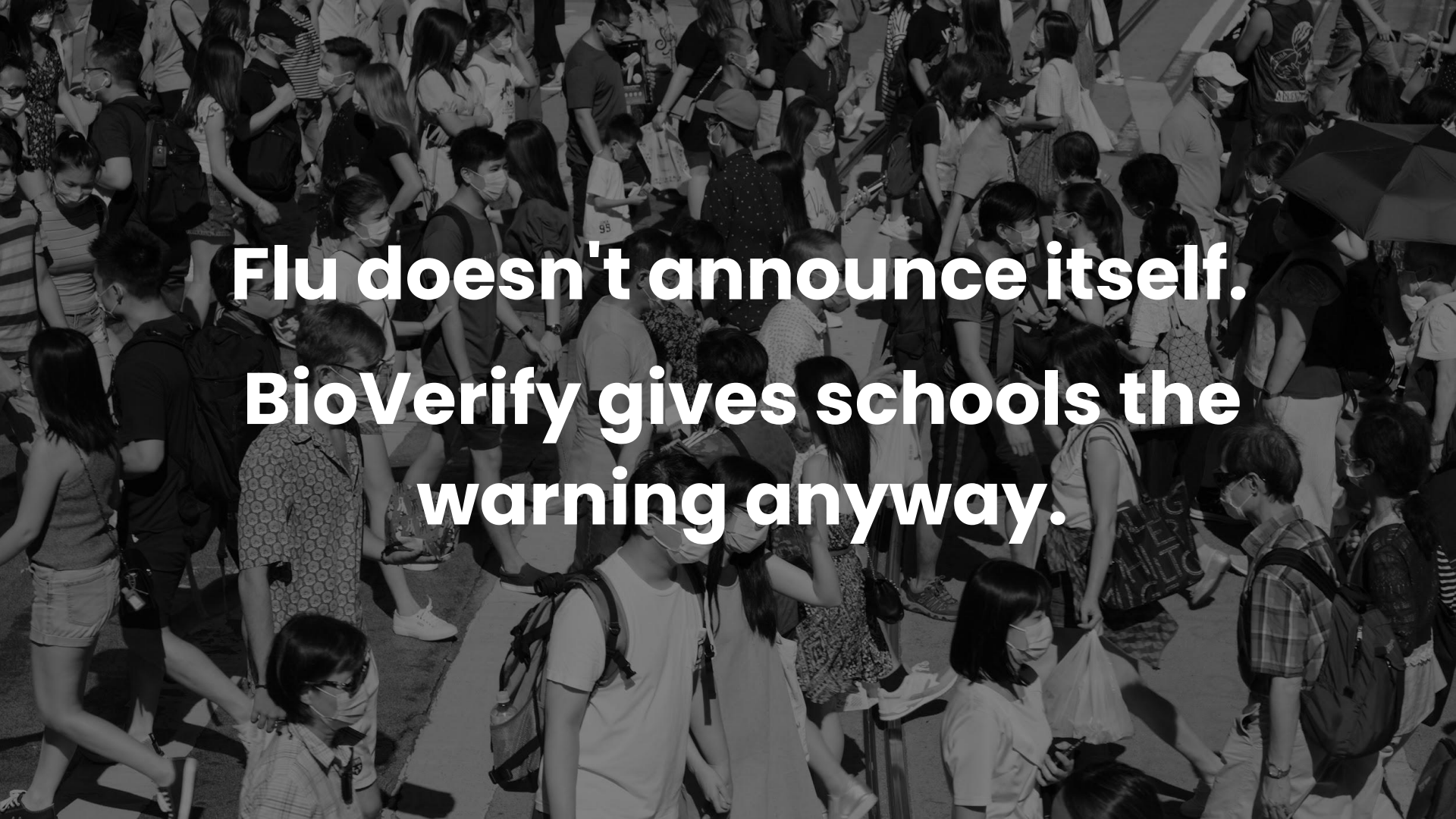
3-5 days

earlier detection

than current surveillance systems

**BioVerify doesn't just protect one school.
It changes how communities respond to flu outbreaks before they start.**

Based on avg. 20-30 infections prevented per school, scaled to 10,000 deployments. Healthcare cost estimate at \$400/infection (CDC).



**Flu doesn't announce itself.
BioVerify gives schools the
warning anyway.**

Meet the Team



Mohamad Amr Idlibi

Data Analysis

Structured and explored sample datasets, identifying key variables relevant to influenza spread and supporting initial model logic.



Uygar Ceylan

Electronics

Led the electronics aspect of the project, focusing on system functionality and integration. developed a hardware concept and workflow for sensor-based data collection within the school setting.



Ivana Okpakowodo

Artificial Intelligence

Built a prototype model to demonstrate how symptom and environmental data could be used to identify potential influenza patterns.



Adham Tarek

Biology

Conducted research on influenza transmission dynamics and symptom progression, ensuring the system's predictions are biologically accurate and contextually relevant.

